

WEEK-4 — BOOKFLOW INTELLIGENT CATALOG & API BLUEPRINT

PART 1 — MySQL Workbench

Step 1 — Create Week-4 Database

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following commands:

```
1 • DROP DATABASE IF EXISTS week_4;
2
3 • CREATE DATABASE week_4;
4
5 • USE week_4;
```

The Output window displays the results of these commands:

#	Time	Action	Message	Duration / Fetch
1	12:26:41	DROP DATABASE IF EXISTS week_4	0 row(s) affected, 1 warning(s): 1008 Can't drop data...	0.000 sec
2	12:26:41	CREATE DATABASE week_4	1 row(s) affected	0.015 sec
3	12:26:41	USE week_4	0 row(s) affected	0.000 sec

Step 2 — Verify Database

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following command:

```
6 • SELECT DATABASE();
```

The Output window displays the results of this command:

#	Time	Action	Message
1	12:26:41	DROP DATABASE IF EXISTS week_4	0 row(s) affected, 1 w
2	12:26:41	CREATE DATABASE week_4	1 row(s) affected
3	12:26:41	USE week_4	0 row(s) affected
4	12:29:38	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned

PART 2 — Create Books Table

Step 3 — Create Books Table

```
7 • CREATE TABLE Books
8 (
9     book_id INT AUTO_INCREMENT PRIMARY KEY,
10    title VARCHAR(200) NOT NULL,
11    author VARCHAR(150) NOT NULL,
12    genre VARCHAR(100),
13    description TEXT,
14    published_year INT
15 );
```

Output

Action Output

#	Time	Action	Message
1	12:26:41	DROP DATABASE IF EXISTS week_4	0 row(s) affected, 1 warning(s)
2	12:26:41	CREATE DATABASE week_4	1 row(s) affected
3	12:26:41	USE week_4	0 row(s) affected
4	12:29:38	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	12:31:03	CREATE TABLE Books (book_id INT AUTO_IN...	0 row(s) affected

Step 4 — Check Table

```
16 • DESCRIBE Books;
```

Result Grid

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(200)	NO		NULL	
	author	varchar(150)	NO		NULL	
	genre	varchar(100)	YES		NULL	
	description	text	YES		NULL	

Result 2 × Read Only

Output

Action Output

#	Time	Action	Message
1	12:26:41	DROP DATABASE IF EXISTS week_4	0 row(s) affected, 1 warning(s):
2	12:26:41	CREATE DATABASE week_4	1 row(s) affected
3	12:26:41	USE week_4	0 row(s) affected
4	12:29:38	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	12:31:03	CREATE TABLE Books (book_id INT AUTO_IN...	0 row(s) affected
6	12:32:04	DESCRIBE Books	6 row(s) returned

PART 3 — Insert Book Data

Step 5 — Insert 10 Books

```
17 • INSERT INTO Books
18   (title, author, genre, description, published_year)
19   VALUES
20   ('The Martian', 'Andy Weir', 'Science Fiction',
21    'An astronaut struggles to survive alone on Mars.', 2011),
22
23   ('Dune', 'Frank Herbert', 'Science Fiction',
24    'A story involving planets, politics, survival and space.', 19
25
26   ('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Scien
27   'A humorous adventure across space and alien worlds.', 1979),
```

Output

Action Output

#	Time	Action	Message
✓ 2	12:26:41	CREATE DATABASE week_4	1 row(s) affected
✓ 3	12:26:41	USE week_4	0 row(s) affected
✓ 4	12:29:38	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓ 5	12:31:03	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected
✓ 6	12:32:04	DESCRIBE Books	6 row(s) returned
✓ 7	12:33:40	INSERT INTO Books (title, author, genre, descript...	10 row(s) affected Records: 10 D

Step 6 — Display Books

```
50 • SELECT
51   book_id,
52   title,
53   author,
54   genre,
55   published_year
```

Result Grid

book_id	title	author	genre
1	The Martian	Andy Weir	Science F
2	Dune	Frank Herbert	Science F
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science F
4	Pride and Prejudice	Jane Austen	Romance

Books 3 x Read Only Con

Output

Action Output

#	Time	Action	Message
✓ 3	12:26:41	USE week_4	0 row(s) affected
✓ 4	12:29:38	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓ 5	12:31:03	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected
✓ 6	12:32:04	DESCRIBE Books	6 row(s) returned
✓ 7	12:33:40	INSERT INTO Books (title, author, genre, descript...	10 row(s) affected Records: 10 Dupli
✓ 8	12:40:56	SELECT book_id, title, author, genre, ...	10 row(s) returned

Step-7 Add Embedding Column

```
59 • ALTER TABLE Books
60 ADD COLUMN embedding JSON;
61
```

Output

Action Output

	#	Time	Action	Message
✓	4	12:29:38	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓	5	12:31:03	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected
✓	6	12:32:04	DESCRIBE Books	6 row(s) returned
✓	7	12:33:40	INSERT INTO Books (title, author, genre, descript...	10 row(s) affected Records: 10
✓	8	12:40:56	SELECT book_id, title, author, genre, ...	10 row(s) returned
✓	9	13:36:06	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0

```
62 • DESCRIBE Books;
```

```
63
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	Field	Type	Null	Key	Default	Extra
	author	varchar(150)	NO		NULL	
	genre	varchar(100)	YES		NULL	
	description	text	YES		NULL	
	published_year	int	YES		NULL	
	embedding	json	YES		NULL	

Result Grid

Result 4

Output

Action Output

	#	Time	Action	Message
✓	5	12:31:03	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected
✓	6	12:32:04	DESCRIBE Books	6 row(s) returned
✓	7	12:33:40	INSERT INTO Books (title, author, genre, descript...	10 row(s) affected Records:
✓	8	12:40:56	SELECT book_id, title, author, genre, ...	10 row(s) returned
✓	9	13:36:06	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0
✓	10	13:36:43	DESCRIBE Books	7 row(s) returned

Step-8 Insert Mock Vector for The Martian

```
UPDATE Books
SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
WHERE book_id = 1;
```

Context Help

Output

Time	Action	Message
12:32:04	DESCRIBE Books	6 row(s) returned
12:33:40	INSERT INTO Books (title, author, genre, descript...	10 row(s) affected Records: 10 Duplicates: 0 Wa.
12:40:56	SELECT book_id, title, author, genre, ...	10 row(s) returned
13:36:06	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0 Duplicates: 0 Warni.
13:36:43	DESCRIBE Books	7 row(s) returned
13:38:03	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ..

Step-9 Insert Mock Vector for Dune

```
67 • UPDATE Books
68 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
69 WHERE book_id = 2;
```

Cont

Output

Action Output

#	Time	Action	Message
7	12:33:40	INSERT INTO Books (title, author, genre, descript...	10 row(s) affected Records: 10 Duplic
8	12:40:56	SELECT book_id, title, author, genre, ...	10 row(s) returned
9	13:36:06	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0 Duplicat
10	13:36:43	DESCRIBE Books	7 row(s) returned
11	13:38:03	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Cf
12	13:38:49	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Cf

Step 10 — Insert Mock Vector for Hitchhikers Guide

```
71 • UPDATE Books
72 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
73 WHERE book_id = 3;
```

Context Help

Output

Action Output

#	Time	Action	Message
8	12:40:56	SELECT book_id, title, author, genre, ...	10 row(s) returned
9	13:36:06	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0 Duplicates: 0 Wan
10	13:36:43	DESCRIBE Books	7 row(s) returned
11	13:38:03	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1
12	13:38:49	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1
13	13:39:40	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1

Step 11--Insert Mock Vector for Pride and Prejudice

```
75 • UPDATE Books
76 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
77 WHERE book_id = 4;
```

Output

Action Output

#	Time	Action	Message
✓ 9	13:36:06	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0 Duplicates:
✓ 10	13:36:43	DESCRIBE Books	7 row(s) returned
✓ 11	13:38:03	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Char
✓ 12	13:38:49	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Char
✓ 13	13:39:40	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Char
✓ 14	13:41:03	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Char

Step 12--Verify All Vectors

```
• SELECT
    book_id,
    title,
    embedding
FROM Books
```

Table Grid | Filter Rows: | Export: | Wrap Cell C

book_id	title	embedding
	The Martian	[0.90, 0.80, 0.20]
	Dune	[0.85, 0.75, 0.25]
	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	Pride and Prejudice	[0.10, 0.20, 0.90]
	The Great Gatsby	[0.15, 0.30, 0.80]

5 x

Action Output

	Time	Action	Message
16	13:42:02	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affe
17	13:42:13	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affe
18	13:42:25	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affe
19	13:42:36	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affe
20	13:42:45	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affe
21	13:43:45	SELECT book_id, title, embedding FROM ...	10 row(s) ret

Step 13-- L2 Distance Search

```
134 FROM Books
135 WHERE embedding IS NOT NULL
136 ORDER BY l2_distance ASC
137 LIMIT 3;
```

Result Grid

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735	
2	Dune	Frank Herbert	Science Fiction	0.0866	

Books 6 Result 7 ×

Output

Action Output

#	Time	Action	Message
✓ 18	13:42:25	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows
✓ 19	13:42:36	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows
✓ 20	13:42:45	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows
✓ 21	13:43:45	SELECT book_id, title, embedding FROM ...	10 row(s) returned
✓ 22	13:45:35	SELECT book_id, title, embedding FROM ...	10 row(s) returned
✓ 23	13:45:35	SELECT book_id, title, author, genre, ...	3 row(s) returned

Step 14---Cosine Similarity Search

```
171 ) AS cosine_similarity
172 FROM Books
173 WHERE embedding IS NOT NULL
174 ORDER BY cosine_similarity DESC
175 LIMIT 3;
```

Result Grid

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986	
8	Enders Game	Orson Scott Card	Science Fiction	0.9983	

Result 8 ×

Output

Action Output

#	Time	Action	Message
✓ 19	13:42:36	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows match
✓ 20	13:42:45	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows match
✓ 21	13:43:45	SELECT book_id, title, embedding FROM ...	10 row(s) returned
✓ 22	13:45:35	SELECT book_id, title, embedding FROM ...	10 row(s) returned
✓ 23	13:45:35	SELECT book_id, title, author, genre, ...	3 row(s) returned
✓ 24	13:46:45	SELECT book_id, title, author, genre, ...	3 row(s) returned

